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Safety in Wearable Robotic Exoskeletons: Design, Control, and Testing Guidelines

Exoskeletons, wearable robotic devices designed to enhance human strength and endurance, find applications in various fields such as healthcare and industry; however, stringent safety measures should be adopted in such settings. This paper presents a comprehensive exploration of challenges associated with exoskeleton technology, ranging from mechanical issues to regulatory and ethical considerations. The enumerated challenges include joint hyper-extension or flexion, rapid or sudden motion, misalignment, fit, and comfort issues, mechanical failure, weight and mobility limitations, environmental challenges, power supply issues, high energy consumption and regeneration, fall risk or stability concerns, sensor failures, control algorithm malfunctions, machine-learning model challenges, communication disconnection, actuator malfunctions, unexpected human-robot interactions, and regulatory and ethical considerations. The paper outlines possible risks and suggests practical solutions based on design, control, and testing methods for each challenge. The objective is to offer a guideline for developers and users, emphasizing safety, reliability, and optimal performance in the ever-evolving landscape of exoskeleton technology. The guideline covers pre-operation checks, user training, emergency response, real-time monitoring, and user interaction to ensure responsible innovation and user-centricity in exoskeleton development and deployment.

Keywords: Exoskeletons, Wearable robotics, Safety assurance, Reliability, Trustworthy

1 Introduction

Wearable robots are systems designed to enhance human physical capabilities by integrating mechanical elements, sensors, actuators, and control systems with end-user biomechanics. Exoskeletons, a specific type of wearable robot, can support and augment limbs, such as the legs or arms, by amplifying muscle strength and endurance through careful mechanical actuation [1,2]. These devices often integrate the proficiency of both humans and machines into their design, to ultimately augment the wearer's capabilities. With careful hardware and controls design, collaborative human-exoskeleton systems can perform motor tasks with greater proficiency level compared to the individual capacities of either isolated component, i.e., the integrated system is greater than the sum of its parts. As a result, the prominence of exoskeleton systems is increasing as a resource to aid individuals with physical disabilities or weakness.

Wearable exoskeleton robots have a multitude of applications that span many areas of interest: medical rehabilitation [3,4], the military [5], industrial and manufacturing sectors [6,7], assistive devices for individuals with mobility disorders [8], sports and athletics, construction and heavy labor industries, space exploration [9], agriculture, emergency response, and even in gaming and entertainment [10]. In the medical field, exoskeletons play a important role in rehabilitation, aiding individuals in relearning walking patterns and rebuilding muscle strength [10]. In the military, these devices enhance soldiers' physical capabilities by assisting with heavy loads as well as reducing fatigue during prolonged missions [11]. Many industries utilize exoskeletons to augment worker strength and endurance, often with the overall goal of minimizing the risk of musculoskeletal (MSK) injuries and improving productivity [6]. As assistive devices, exoskeletons empower individuals with mobility challenges, providing increased independence in daily living activities [8]. In construction, and agriculture, exoskeletons contribute to improved performance, reduced physical strain, and enhanced efficiency. Lastly, in space exploration, these devices offer additional support to astronauts in navigating micro-gravity environments [9].

Despite their usefulness and growing demand, substantial challenges remain in the deployment of exoskeletons: effective implementation of human-robot interaction (HRI), mechanical designs, and development of control systems [12]. On top of these, the safety of exoskeletons is foremost, considering their potential to considerably enhance human physical capabilities [13]. Ensuring the well-being of users is of utmost importance [6,14]. As such, exoskeletons must undergo rigorous safety assessments to mitigate the risk of injuries. The mechanical components, sensors, and control systems demand meticulous engineering to prevent malfunctions or unintended movements that could pose a threat to the wearer. Moreover, the interface between the exoskeleton and the human body requires careful consideration to avoid pressure points, discomfort, or potential MSK issues [6]. Despite their numerous benefits, it is essential to acknowledge that injuries may still occur. This fact motivates much research and development centered around addressing said safety concerns, refining design features, and establishing comprehensive guidelines for the safe deployment and usage of exoskeleton technology [15,16]. Balancing innovation with safety measures is crucial to harness the full potential of exoskeletons.

Despite the rapid advancement of exoskeleton technology, there remains an absence of guidelines regarding the safe deployment of exoskeleton robots [13,17]. Regrettably, current review papers have addressed the significance of safety in a few sentences [2,7,10,12,18–23], or entirely overlooked it in most cases. To the best of our knowledge, this review paper is the first to focus specifically on the aspect of safety. The field currently requires a set of standards and protocols that address safety concerns systematically. The current article highlights the complexity of ensuring the safety of users, emphasizing the need for careful considerations in mechanical design, control systems, and complex HRI (Figure 1). As the demand for exoskeletons continues to grow, a standardized framework for safety becomes imperative [16,17]. Establishing comprehensive guidelines would not only serve to protect users from potential injuries but also foster confidence in the broader adoption of exoskeleton technology [15]. It is essential for researchers, engineers, and regulatory bodies to collaborate in developing a robust and universally applicable set of safety guidelines that can guide the safe deployment of exoskeleton robots across

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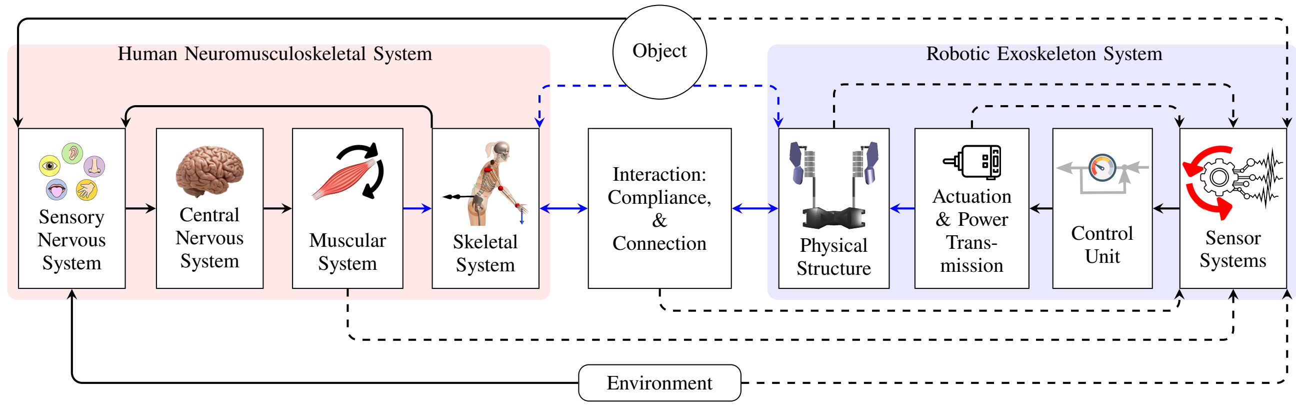


Fig. 1 The diagram of HRI structure. The force/torque and the signals are in blue and black, respectively. The dashed lines may connect or not, based on the situation of object handling, installed sensors, type of sensors, and place of sensor installation.

various contexts.

Previous review papers have focused on application, features, performance metrics, and comparative analysis of various exoskeletons, primarily examining factors such as speed, torque, weight, autonomy, and user interaction [2,7,10,12,18–23]. However, safety aspects of exoskeletons have been largely overlooked, a gap that we address in this manuscript. By mentioning these critical safety concerns, our study aims to list the gaps and challenges in the current research on exoskeleton adoption and development. The sections of this paper provide a comprehensive exploration of challenges and potential solutions in the implementation of exoskeleton technology (section 2) in addition to an in-depth analysis of mechanical, regulatory, and ethical considerations. Section 3 offers detailed guidelines for developers and users, emphasizing safety, reliability, and optimal performance in exoskeleton development. Following this, section 4 presents two relevant case studies showcasing successful implementation of safety measures, both in simulation and experiment. Lastly, section 6 summarizes key insights on responsible innovation and user-centric development of exoskeletons across diverse applications.

2 Challenges and solutions

Within this section, issues and injurious events pertinent to exoskeletons are discussed, followed by the delineation of solutions related to the design, control, and testing of the exoskeleton. The objective herein is to offer comprehensive measures to ensure the safety of the exoskeleton system.

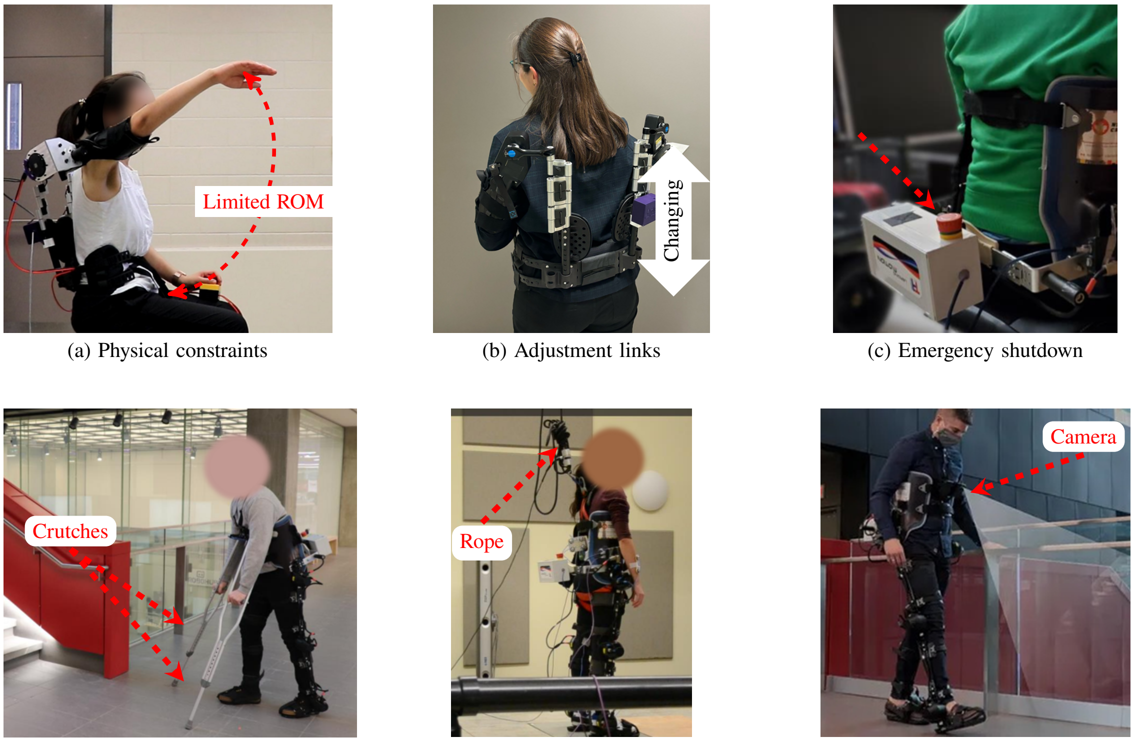
2.1 Joint hyper-extension or flexion. The potential for rotations in exoskeleton joints exceeding 360 ° is a major concern in the deployment of exoskeletons. Hyper-extension or flexion of human joints induced by the device is a notable hazard for the end-user. This is especially the case for individuals like older adults, who experience age-related decreases in passive range of motion (RoM) [24].

Addressing this concern involves the implementation of numerous precautionary measures that mitigate the risks associated with unrestricted joint motion. Notably, mechanical stops are incorporated into the design to limit the RoM (Figure 2(a)), as suggested by existing research [8,25]. Additionally, electronic sensors, specifically limit switches [14], and software controls are integral components aimed at preventing hyper-extension or flexion of joints [3]. The consideration of the RoM for both upper-limb [26] and lower-limb [27] applications further informs this approach. The amalgamation of these precautionary measures contributes to a secure biomechatronic system.

2.2 Rapid or sudden motion. High acceleration, deceleration, or velocity of exoskeletons can induce undue stress or harm to the user's joints, muscles, and other anatomical structures, as evidenced in the literature [28]. It is imperative to address and mitigate the potential for kinetic injuries during the design and operation of exoskeletons to safeguard user safety and overall well-being. One viable approach involves implementing constraints on the maximum speed of exoskeleton joints within the control loop, aligning them with human peak joint velocity (upper-limb [26] and lower-limb [27]). To further enhance safety measures, offline testing of the exoskeleton, as proposed in [28], becomes instrumental in identifying potential issues and refining the system's performance. Additionally, the measurement and subsequent restriction or minimization of jerk (the derivative of acceleration) promote smoother and more comfortable movements within the exoskeleton system [29]. However, there are challenges associated with obtaining reliable jerk estimates, for example, from numerical differentiation of accelerometer signals or a model-based observer. Regardless, these collective solutions offer a comprehensive approach to addressing kinetic injury risks during exoskeleton operation.

2.3 Misalignment, fit, and comfort. Challenges associated with misalignment, fit, and comfort in exoskeletons give rise to diverse issues encompassing discomfort, fatigue, restricted mobility, and skin irritation [30]. Ill-fitting exoskeletons can lessen a user's capacity to don the device for prolonged periods as they reduce both comfort and usability [31]. The emergence of pressure ulcers and skin irritation poses a significant concern, potentially escalating to more serious health implications [32]. Furthermore, an uncomfortable exoskeleton has the potential to compromise overall performance, particularly in tasks demanding precision and agility; the end-user may adopt motor patterns that limit discomfort instead of focusing on executing the task. In this case, poor user satisfaction results in further disuse of the device.

Mitigating the challenges associated with misalignment, fit, and comfort necessitates a strategic focus on customization, ergonomic design, and the utilization of comfortable materials with sufficient padding [33]. The incorporation of adaptive components for customization, adherence to ergonomic design principles, and the integration of breathable materials represent viable solutions to augment the fit and comfort of exoskeletons [34]. Design strategies involving redundant joints or self-aligning joints [5] in exoskeletons can facilitate alignment with intricate human joints, such as the shoulder [35] or knee [36]. The systematic gathering of user feedback through iterative design processes and biomechanical analyses ensures ongoing refinement grounded in real-world experiences. Additionally, comprehensive user training encompassing proper exoskeleton-use and adjustments is important in optimizing fit and comfort (Figure 2(b)).



(a) Physical constraints (b) Adjustment links (c) Emergency shutdown

(d) Crutches or walker (e) Body weight support (f) Environmental perception

Fig. 2 Examples of present safety mechanisms in exoskeleton technology

Ergonomic design (e.g., ISO 9241) is the scientific discipline concerned with designing products that optimize human well-being and overall system performance. This involves considering the physical interactions between the user and the device to minimize discomfort, fatigue, and risk of injury [37]. In the prototyping stage, ensuring an ergonomic design can involve both qualitative and quantitative assessments. Qualitatively, this includes user feedback and expert reviews on the comfort and usability of the device. Quantitatively, metrics such as joint angle ranges, pressure distribution, and muscle activation levels can be measured and incorporated into a design optimization algorithm [35]. Regarding control, measuring the achievement of an ergonomic posture involves assessing how well the exoskeleton supports natural and comfortable body movements. This can be quantified through metrics such as alignment of the exoskeleton's joints with the user's anatomical joints, the degrees of freedom (DoFs) in movement, and the forces exerted on the user's body. Compliance between the exoskeleton and the user is crucial, as it affects both the comfort and effectiveness of the device. Compliance losses, which occur when there is a mismatch between the user's movement and the exoskeleton's response, should be minimized to ensure optimal performance. These losses can be measured using sensors that track the relative motion and force exchange between the user and the exoskeleton.

The maximal torque generation with isometric, isokinetic, and passive conditions, focusing on the upper body are documented in [26], while corresponding data for the lower body are covered in [27]. Although human joint torque levels can be increased by the exoskeleton, excessive increments have the potential to induce discomfort. The amount of pressure that leads to pressure points and discomfort on human skin varies based on factors such as the body area involved, duration of pressure, individual sensitivity, and skin condition [38]. To prevent discomfort and skin issues, it is crucial to distribute pressure evenly, use materials that minimize friction, encourage movement to relieve pressure, and adjust pressures based on individual needs and comfort levels.

2.4 Mechanical failure. Over the course of any biomechatronic device's lifespan, components such as joints, actuators, and/or structural elements undergo wear and tear. These small (or large) changes in hardware integrity culminate in breakage(s) or malfunction(s) that compromise exoskeleton function [20,39]. Failures may arise due to prolonged usage of particular components, surpassing design limitations (repeated or acute), or unexpected stresses encountered during operation. High stresses can be of notable concern, especially in the cases of soft exo-suits over rigid exoskeletons, which can experience larger strains prior to failure and fewer issues like joint misalignment [40,41]. Given the potential ramifications for user safety, addressing and mitigating the risks associated with mechanical failures is paramount.

A primary resolution for this concern involves designing the exoskeleton with robust materials and subjecting it to meticulous computer-aided engineering analysis in early stages [20]. The judicious integration of high-quality materials and resilient design principles in critical components serves to augment durability, reinforcing the exoskeleton against mechanical failures that pose a threat to user safety. Additionally, the adoption of a systematic maintenance regimen, coupled with routine inspections, emerges as a pivotal strategy. This procedural framework facilitates the early identification of wear or potential issues, thereby facilitating prompt interventions such as repairs or component replacements. Moreover, the incorporation of real-time monitoring systems and sensors ensures continuous assessment of the mechanical components' condition. In tandem, the strategic implementation of emergency shutdown mechanisms (Figure 2(c)), e.g., easily accessible while donned, offers a swift response to potential or detected failures, introducing an additional layer of protective measures.

2.5 Weight and mobility limitations. The weight and rotational inertia of the exoskeleton influences the end-user's mobility, agility, and overall motion [42]. As such, there is the potential for MSK injuries to develop. More specifically, the substantial weight of exoskeletons may induce discomfort, fatigue, and limitations in mobility. Sometimes, the weight distribution of the system, particularly due to the high weight of the power source [43] and actuators, contributes to discomfort and feelings of immobility [44].

Critical considerations to address device weight include the selection of lightweight yet robust materials (such as, aluminum alloy Al7075, carbon fiber [45], or polyether ether ketone [46]), the infusion of ergonomic design principles into the device, and the methodical distribution of the exoskeleton's weight with respect to the user's body [42]. The incorporation of passive mechanisms, in conjunction with lighter and smaller actuators specifically tailored for optimal functionality, plays an important role in fostering the creation of a lightweight exoskeleton [35]. Nasr et al [35] introduced an exoskeleton featuring a novel passive-active actuator. This innovative design is not only lighter but also capable of producing the necessary torque through passive mechanisms as well as a brushless direct current (BLDC) motor. Overall device inertia can be further reduced through the incorporation of soft materials throughout the device, i.e., exosuits with cable actuation [40,41]. Strategic weight reduction can also be achieved through the pragmatic placement of the heavy power supply on the ground and the judicious tethering of the exoskeleton; however, these are only suitable for controlled environments. A further avenue for weight reduction is explored through the design of an electric regenerative exoskeleton [47]. Notably, the back electromotive torques can dampen gravitational torques acting at separate joints, though this can also be achieved through model-based torque assistance [48]. We note that joint torque-level compensation with gravitational loads can only do so much due to the under-actuated nature of bipedal dynamics.

Rigid exoskeletons offer high structural support and load-carrying capacity, suitable for heavy lifting and industrial applications, but they lack flexibility and can restrict natural movement [2]. Soft exoskeletons provide flexibility and comfort, ideal for assistive and rehabilitation purposes, but they lack support for heavy loads and may have durability issues [9]. Rigid-soft hybrid exoskeletons combine the strengths of both systems, offering structural support and flexibility, but are complex in design [49]. Effective weight distribution is crucial in exoskeleton design, with serial linkage mechanisms being simple and effective for precise positioning but potentially causing high stress on proximal joints, while parallel mechanisms evenly distribute loads and offer higher stiffness but are complex and bulky [50,51]. Serial-parallel mechanisms balance force distribution and stability but are complex to design and control [52]. Upper body exoskeletons, using lightweight composites and serial linkages [4], enhance arm and shoulder movements, while lower body exoskeletons, employing robust materials and parallel mechanisms, assist with walking and load-carrying, providing support and stability [50].

2.6 Environmental challenges. Exoskeletons can encounter formidable challenges when subjected to adverse environmental conditions. In the presence of extreme temperatures, adverse weather conditions (e.g., rain or snow), and uneven terrains, mechanized suits may experience diminished performance. The intricate electronic and mechanical constituents of exoskeletons are also susceptible to impacts, potentially jeopardizing their operational functionality and safety. For example, extreme temperatures (or extremes of pH or alkalinity) may detrimentally influence the efficacy of sensors and actuators. Additionally, the operation of BLDC motors, known for their substantial Joule heating, necessitates the incorporation of heat sinks to dissipate thermal energy and prevent user discomfort or injury [53]. Furthermore, dusty and wet conditions pose a notable concern as they may compromise the integrity of onboard electronic components, e.g., introducing static electricity and corrosion [7].

Guaranteeing the dependability and feasibility of exoskeleton technology within diverse environmental settings requires a proactive design approach. During the design phase, scrutiny and analysis of the actuator's heat dissipation characteristics and performance under different temperatures is imperative [1,53]. To preserve the system's integrity, protective measures should be incorporated to safeguard the electronic components and sensors from potential hazards [54]. For example, adequate insulation of components to limit contact with air, dust, and water is necessary; however, one must simultaneously be considerate of how such features may affect the internal temperature and weight.

2.7 Power supply issues. The power sources employed in active exoskeletons, whether in the form of batteries for portable configurations or connections to electrical grids for tethered counterparts, pose inherent hazards to the user [43]. It is important to prioritize user safety in these cases, as injuries stemming from power-related issues can be deadly. Safety and performance standards for electrical and electronic equipment should be met prior to in-person use. These standards are typically established in accordance with local and/or national governing bodies, e.g., CSA and/or UL certifications. Designing active exoskeleton robots with such certifications further instills confidence in users regarding the reliability, functionality, and safety of the technology during operation. Notably, the batteries utilized in these exoskeletons carry the risk of swelling or explosion under specific circumstances. Lithium-ion batteries, commonly integrated into diverse electronic devices, are susceptible to issues such as thermal runaway, wherein uncontrolled internal temperature escalation may result in release of gas, expansion or rupture of the battery casing, and, in extreme scenarios, combustion or explosion [55]. Simultaneously, in systems tethered to electrical grids, the prospect of a short circuit between the exoskeleton and the human operator necessitates careful consideration [56].

Manufacturers must proactively integrate safety features and design considerations to mitigate risks associated to the power supply. This approach involves the incorporation of built-in thermal management systems, protective circuitry, and the use of durable casing materials [55,57]. Additionally, grounding the exoskeleton body becomes imperative for human safety in the context of electrical grid connections [56]. Rigorous testing protocols and strict adherence to safety standards are indispensable to ensure the secure functionality of exoskeletons and preempt battery-related incidents [57]. Users also play a pivotal role in maintaining safety by adhering to recommended charging procedures, avoiding overcharging, and promptly reporting any observed abnormalities to the manufacturer or relevant authorities, e.g., lithium-ion battery swelling. This multifaceted approach underscores the comprehensive commitment to safeguarding the well-being of individuals engaged with active exoskeleton technology.

2.8 High energy consumption. To meet the demand for optimal joint-level performance, i.e., high torque capacity with low inertia, a direct drive (or quasi-direct drive) BLDC motor is generally employed in active exoskeletons [21,35,58]. However, using BLDC motors comes with several caveats. To elaborate, BLDC motors have high electric consumption [58], which can result in increased overall energy costs. On a larger scale, such wasteful components can contribute to environmental concerns due to their large power requirements (i.e., large ecological footprint) and their associated supply. Elevated energy consumption can lead to higher operational expenses, particularly in applications where these motors are extensively used.

One effective solution to this concern is the implementation of regenerative systems [21,58]. Regenerative systems are designed to recover and reuse energy that would otherwise be dissipated as heat during braking or deceleration. In the context of BLDC motors, regenerative braking systems can be incorporated to capture the work generated during braking and convert it back into electrical energy [58], i.e., the motor is temporarily considered a generator. This recovered energy can then be fed back into the power supply or stored for later use.

2.9 Reduced upright stability and fall risk. Donning a lower-limb exoskeleton is often associated with reductions in the capacity to regulate upright balance [22,48,59–61]. A notable risk includes the potential for falls, which, in turn, may lead to fractures, sprains, or other MSK injuries [13,22]. Impact(s) from a fall can also damage the exoskeleton hardware resulting in expensive repair/replacement. The compromise in stability may emanate from factors such as sensor inaccuracies, mechanical malfunctions, insufficient joint torques, limited mechanical DoFs, and increased actuator impedance (rigidity) [25,60,61]. Factors like inappropriate weight distribution or unanticipated environmental variations may further contribute to the danger of compromised stability [22,61]. Moreover, falls have the potential to incur psychological effects that impact user confidence and trust in the exoskeleton technology as well as evoking a fear of falling.

Improving the upright stability of lower-limb exoskeletons is most often achieved by introducing a walker or crutches and/or a ceiling-mounted tether. The inclusion of a walker or crutch (Figure 2(d)) provides valuable support aid, offering users additional points of contact with the ground to increase the support polygon size and enhance stability, particularly in challenging terrains [23]. The downside of this solution is the limits placed on arm movement/use, i.e., end-user must prioritize arm-use for stability rather than operations in a workspace. Simultaneously, the integration of a tether and harness (Figure 2(e)) provides a suspension/support system that can assist in weight bearing and act as a safety net in the event of a loss of balance; however, these setups are better suited for controlled environments like a laboratory as opposed to real-world environments. Alternatively, some exoskeletons are equipped with airbags to protect patients from injuries in the event of a fall [40].

Addressing stability concerns and mitigating fall risks in lower-limb exoskeletons suitable for the real world necessitates a comprehensive approach incorporating technological innovations, design enhancements, robust control algorithms, and user-focused strategies. One important notion involves the integration of advanced sensor technologies, such as inertial measurement unit (IMU) and force sensors, to elevate the exoskeleton's capacity for precise movement detection and responsive feedback. Such systems can be used for explicit disturbance detection to trigger feed-forward assistive torque assistance [62,63]. In other instances, assistance in balance regulation is delivered through some form of model-free or model-based feedback control based on information from available sensors. These systems are typically designed to assist the end-user in regulating system balance in the case of walking [59,62,64–66] or standing [48,61,63,67], but not necessarily both. Implementing assist-as-needed balance control schemes can be challenging, especially if the desire is to reduce user "slacking", i.e., cases where the user relaxes and allows the device to contribute as much torque as possible to the task [66,67]. Recent simulation and experimental research into time delays with human-exoskeleton balance suggest that assistive torques should be generated prior to the user's reaction [61,63,68]. For instance, one can use surface electromyography (sEMG) signals (muscle activity, which are measurable before the actual muscle contraction and subsequent movement occurs, or visual environment perception (such as cameras) to detect human motion, as investigated by Laschowski et al [69]. Therefore, relying on users to initiate balance recovery within an assist-as-needed framework may not be particularly advantageous. Additionally, the utilization of cameras for environmental perception (Figure 2(f)) and proactive planning to avoid hazards in undetected surroundings has the potential to bolster exoskeleton fall prevention capabilities [69].

As a final note, mechanical considerations also play an important role in addressing stability challenges in lower-limb exoskeletons. Designing these robots with mechanical redundancies, such as redundant actuators or adaptable joint mechanisms, can yield fail-safe mechanisms that compensate for unexpected failures. Additionally, optimizing weight distribution across the exoskeleton could help improve the integrated human-exoskeleton's capacity to balance by repositioning the whole-body center of mass within more advantageous areas of the feasible stability region during operation [61,70]. Inclusion of high-level environmental sensing capabilities may further enhance stability by allowing the exoskeleton to adapt to diverse terrains and unforeseen obstacles. Integrating user training protocols, emergency stop mechanisms, and feedback systems provides a user-centered approach to stability. User training programs educate individuals on proper operation and response procedures, while emergency stop mechanisms and feedback systems offer immediate insights and control, reducing the likelihood of falls and enhancing overall user safety.

2.10 Sensor failures. Active exoskeletons rely on sensors to measure various states (e.g., joint angles, speed, and forces); however, the functionality of these exoskeletons can be compromised due to sensor malfunctions and noise, especially in the case of feedback control. One prevalent issue pertains to sensor white or high-frequency noises characterized by small amplitudes [71]. For example, IMUs experience white noise and their computed angle has drift resulting from integration of measured acceleration noise [72]. Another concern involves random responses, and a third challenge is sensor disconnection [73,74]. These issues pose a significant risk, potentially resulting in catastrophic consequences if the control loop is reliant on the raw sensor values and not prepared to address noise.

To mitigate these challenges, a robust control system that do not rely on measurements alone can implemented. Specifically, measures should be implemented to address each identified issue comprehensively. For instance, the issue of high-frequency, low-amplitude noises can be effectively rectified by integrating a low-pass filter into the system [71], i.e., improving the signal-to-noise ratio. It is noteworthy that human motion tends to be characterized by a comparatively slower pace than that of high-speed robotic motions [75]. Consequently, the employment of a low-pass filter emerges as a judicious solution [76], though it may come at the cost of introducing phase shift within the measured signal. An alternative, lag-free solution can be obtained by using a model-based state observer [77].

The problems associated with sensor disconnection during operation can be proactively managed through the implementation of redundant sensor connections or a sensor-health monitoring system. A sensor health monitoring system continuously evaluates the condition of sensors, promptly identifying and addressing any anomalies. Redundant sensors in conjunction with sensor fusion algorithms [77,78] can minimize the impact of a single sensor failure, similar to models of the central nervous system (CNS) [79]. Sensor fusion using a model-based state observer can also help validate consistency of sensor data with the current state of the control system's internal model. The internal predictive model anticipates the expected sensor measurements based on the system's dynamics, previous states, and the commanded control input [74,77,80]. By comparing the actual sensor values with the predicted values, deviations can be identified and corrected. If the current sensor measurements fall outside the acceptable range determined by historical or average values, it can also be flagged as an anomaly. This enables the system to reject or abstain from using unreliable data, preventing potential adverse effects on exoskeleton performance.

2.11 Control algorithm malfunction. Active exoskeletons heavily depend on sophisticated control systems to manage precise and safe movements. Any anomalies in the software or hardware of the control system can result in unpredictable motions, potentially giving rise to safety hazards.

To ensure the reliability of the control algorithms, a comprehensive testing protocol is important. Initially, the control system may undergo rigorous checks and simulations using a virtual model that closely mimics human biomechanics and the associated hardware. This virtual testing phase aims to preemptively identify and rectify any issues before the exoskeleton is employed with a real user [81]. The human model can be a neuromusculoskeletal system controlled be the CNS [82]. It is essential that the control system exhibits seamless functionality in this simulated environment, demonstrating its capability to govern movements accurately and safely. Furthermore, in simulation, elements of stochasticity as well as unanticipated disturbances can be included to assess robustness, i.e., signals should contain some noise and some disturbance force/torque should be applied to the exoskeleton frame.

Control systems should also be subjected to a phased testing approach when possible, progressing from a mannequin to a real human user. This staged testing strategy allows for a gradual assessment of the exoskeleton's performance, ensuring a controlled and monitored transition to human interaction [83]. The entire RoM and functional capabilities of the exoskeleton must undergo meticulous scrutiny during these testing phases. This encompasses evaluating joint movements, response times, and the overall coordination of the exoskeleton with the user's actions.

2.12 Inaccurate machine-learning models. Instead of mathematical models, control algorithms occasionally employ machine-learning models. An illustrative instance involves their application in interpreting human muscle activation, as measured by sEMG, for prosthetic and exoskeletons control [84]. These contemporary black-box machine-learning models, while gaining traction, are often trained on limited datasets and may not undergo rigorous validation. Their confidence is primarily derived from exact training input-target sets, a circumstance that may not faithfully represent the complexities encountered in practical implementations. Consequently, the generalizability and reliability of such models in real-world scenarios remain subject to scrutiny and necessitate comprehensive validation processes to ensure their efficacy and applicability beyond the training data [85].

In fortifying the confidence and reliability of machine learning models, particularly in the intricate domain of control algorithms for applications like prosthetics and exoskeletons, a holistic testing strategy spanning both offline and run-time scenarios is indispensable. In the offline realm, rigorous cross-validation techniques should be employed to meticulously evaluate the model's performance across various subsets of the training data, thereby discerning its generalizability and mitigating potential over-fitting [86]. Additionally, dataset augmentation becomes important, facilitating exposure to a more diverse range of scenarios and augmenting the model's adaptability to real-world nuances. Hyper-parameter tuning, involving systematic exploration of different configurations, further refines the model's performance. Robustness testing, encompassing the introduction of perturbations or variations to simulate real-world uncertainties, offers insights into the model's resilience under unpredictable conditions.

Transitioning to run-time application, real-time performance monitoring is imperative, ensuring continuous assessment of the model's behavior in dynamic environments. Adversarial testing becomes crucial, subjecting the model to intentional disturbances to evaluate its response under challenging conditions and uncover potential vulnerabilities [85]. Thorough examination of the model's reaction to edge cases and outliers enhances its robustness and applicability in diverse scenarios. As an example, in case of using a classification learning-based model, the Gini Index can be used to check the confidence of the classification [87]. Continuous learning mechanisms enable the model to adapt to evolving conditions, while user feedback integration provides insights into real-world performance and areas for refinement. This comprehensive testing framework, spanning both offline testing [88,89] and run-time contexts [85], ensures that machine learning models not only excel in controlled environments but also exhibit resilience and adaptability when confronted with the complexities of practical, real-world applications.

2.13 Communication or sensor disconnection. Communication breakdowns among distinct components or modules within the exoskeleton system can promote drastic consequences. The absence of seamless coordination between components may result in erratic behavior, where distant sections of the exoskeleton fail to synchronize effectively. Such dysfunction threatens the overall performance and safety of the exoskeleton and user, potentially compromising the user's experience and introducing operational risks.

To address these challenges, robust communication strategies should be implemented. Employing advanced communication protocols establishes reliable and efficient data exchange between components. Concurrently, integrating error-checking mechanisms within the communication framework acts as a safeguard, allowing the system to identify and rectify potential discrepancies promptly. Furthermore, the overall system design should be adept at mitigating communication delays or losses. This involves the incorporation of mechanisms to handle interruptions or latency, ensuring a resilient communication infrastructure that fosters seamless coordination among exoskeleton components. For example, using the industrial controller area network (CAN), as a proven and effective vehicle CAN standard, is advised for wearable robots [90] since it is robust to noise. By emphasizing these measures, the reliability and stability of the communication network within the exoskeleton system can be significantly enhanced, fortifying its overall functionality and user safety.

2.14 Actuator malfunctions. Malfunctions in the actuators, which play a main role in generating movement and providing load support, can be dangerous. Such failures may manifest as compromised movement capabilities, diminished load-bearing support, unforeseen motions, or excessive external wrench (force/torque). The latter poses substantial risks to the safety and efficacy of the exoskeleton.

To mitigate the potential ramifications of actuator failures, a multifaceted approach is imperative. This involves the integration of redundancies within the actuator systems to establish alternative pathways for movement generation and load-bearing functions, e.g., using a passive mechanism (generate fixed torque profile without energy consumption) parallel to an actuator [35]. To address scenarios where the actuator might generate higher joint torque than the desired value, it is crucial to implement controls that limit the maximum torque. This limitation can be achieved through both the control algorithm and direct or indirect measurements integrated into the electrical driver system. In addition, the implementation of fault detection algorithms serves as an active surveillance mechanism, immediately identifying any irregularities in actuator performance and facilitating prompt corrective action. Additionally, a proactive regimen of regular maintenance and systematic testing of actuators is essential to preemptively detect and address any signs of wear, fatigue, or impending malfunctions.

2.15 Unexpected human-robot interaction. Despite restrictions imposed on the velocity and RoM of an exoskeleton, unforeseen motions or unprecedented HRIs [12], e.g., where the exoskeleton robot suddenly provides more torque than necessary to a novice user or resists user intent, can still lead to negative outcomes, such as unexpected stress resulting in potential tissue strain, and the exertion of muscles against the exoskeleton subsequently inducing fatigue and tissue strains. There is a notable risk of injuries to vulnerable soft tissues like tendons and ligaments, particularly if the exoskeleton imparts forces in unintended directions.

To mitigate these challenges, a comprehensive solution is recommended. User training programs focusing on exoskeleton operation and awareness of potential risks play an important role in significantly reducing the likelihood of accidents. For example, a vocal feedback system can be used to inform the user of risky postures/maneuvers before any task has initiated. Additionally, the incorporation of predictive simulation testing for the exoskeleton, as suggested in studies such as [82,91], coupled with the measurement and examination of exoskeleton motions utilizing a simulation of the CNS [81], contribute to an understanding of the system's dynamics and facilitate more comfortable movements.

For instance, Nasr et al [81] observed that enhancing the exoskeleton's strength incrementally is advisable, given that humans may not have adapted their motor patterns. Furthermore, these simulations can also be used to try and assess worst-case scenarios, e.g., when the exoskeleton and CNS have no knowledge of each other [61].

2.16 Regulatory and ethical considerations. Navigating regulatory requirements and addressing ethical concerns in the realm of exoskeleton technology pose a complex challenge for both developers and users. Developers must grapple with evolving regulatory frameworks that vary across regions, requiring a nuanced understanding to ensure compliance with safety, quality, and performance standards. The ethical perspective, particularly regarding liability and consent in the event of malfunctions or accidents, adds another layer of complexity, especially in shared spaces. Establishing clear liability frameworks becomes crucial to ensure accountability and protect the interests of both developers and users.

To address these challenges, a proactive and collaborative approach is essential. Developers can engage in ongoing dialogues with regulatory bodies to foster a better understanding of exoskeleton technology and work towards balanced regulatory frameworks. Transparent communication is challenging, with a focus on informing users about the device's capabilities, limitations, and potential risks. This includes implementing clear and comprehensive consent processes to enable users in making informed decisions and building trust in the technology. Moreover, the development of standardized liability frameworks, in collaboration with legal experts and stakeholders, is crucial for providing clarity in unforeseen circumstances. Overall, a comprehensive strategy involves engaging with regulators, transparent and readily-available communication with users, and establishing clear rules to encourage responsible innovation while prioritizing safety, ethical treatment, and informed consent.

3 Guidelines

In the ever-evolving landscape of exoskeleton technology, ensuring the safety, reliability, and optimal performance of these systems is challenging. This comprehensive guideline addresses key facets of exoskeleton development, spanning from environmental considerations to operational intricacies. The guideline is structured into distinct sections, each focusing on specific critical aspects (Figure 3). By assessing environmental conditions, identifying potential hazards, formulating precise requirements, and incorporating safety-oriented design principles, developers can enhance the robustness and user-centricity of exoskeleton systems. The guideline concludes with a set of questions to guide the operational phase, emphasizing pre-operation checks, user training, emergency response, real-time monitoring, and HRI. Through adherence to these guidelines, the development and deployment of exoskeletons can be executed with suitable consideration for user safety and system efficacy.

3.1 Conditions and application. The environmental resilience of exoskeletons takes base stage, with the following examination:
- Assess the environmental conditions exoskeletons may encounter. Include factors like temperature, humidity, and terrain variations.
- Evaluate the specific applications and tasks the exoskeleton will be used for.
- Consider the range of activities and operational contexts.

3.2 Hazards and loss. A thorough hazard analysis is undertaken, delving into potential risks arising from mechanical failures, sensor malfunctions, and unexpected motions. Briefly, the following steps should be taken:
- Conduct hazard analysis to identify potential risks. Aspects such as mechanical failures, sensor malfunctions, and unexpected motions should be given special attention.
- Systematically document and categorize identified hazards.
- Analyze potential losses associated with identified hazards.
- Categorize losses based on severity and impact on user safety, e.g., personal injury, damage to equipment and/or reputation, financial loss, or legal consequences.
- Establish a hierarchy of criticality that addresses losses accordingly and guides mitigation strategies.

3.3 Requirement analysis. A requirement analysis is strongly recommended in deploying robotic assistive devices. The goal herein is alignment of functional requirements with user needs and operational goals while ensuring safety requirements are inline with regulatory standards. The following action items are recommended:
- Specify the desired tasks and activities that the intended exoskeleton will support.
- Define the functional and safety requirements of the exoskeleton system. Then align functional requirements with user needs and operational goals. Some examples of such requirements include the following: ensuring the exoskeleton can be customized to fit users of various body types comfortably and designing the exoskeleton to prevent excessive force application near user joints.
- Develop safety requirements based on hazard analysis and loss assessment (see 3.2).
- Ensure safety requirements are measurable, achievable, and aligned with regulatory standards.
- Incorporate and document fail-safe mechanisms and emergency procedures.

3.4 Safety-oriented design. This portion of the proposed wearable exoskeleton guidelines focuses on the mechanical, electronic, and ergonomic aspects of exoskeleton development, i.e., where in the physical design can safety measures be incorporated. The following actions are recommended:
- Implement mechanical stops to limit joint RoM and prevent hyper-extension or flexion relative to expected user RoM.
- Integrate redundant actuators and adaptable joint mechanisms as fail-safe components that can help ensure stability.
- Use lightweight yet robust materials for optimal weight distribution.
- Include limit switches and sensors to prevent rapid or sudden motions that could possibly harm the user.
- Employ filters to mitigate sensor noise and enhance reliability. Low-pass filters can be employed, though at the expense of introducing phase shift. Alternative solutions include model-based observers that can remove measurement noise without introducing lag [77].
- Implement redundant sensor connections and sensor health monitoring systems.
- Prioritize customization through adaptive components for addressing fit and comfort issues.
- Infuse ergonomic design principles to minimize misalignment and discomfort.
- Incorporate breathable materials with sufficient padding to mitigate skin irritation.

3.5 Safety-oriented operation questions. Critical aspects such as pre-operation checks, user training, emergency response protocols, real-time monitoring considerations, and HRI and comfort should be systematically addressed. These questions guide developers and operators in ensuring that all essential facets of exoskeleton operation, from systematic checks to user awareness and real-time monitoring, are thoroughly considered for enhanced safety and optimal user experience.
- Are all sensors functioning properly, and have they been calibrated?

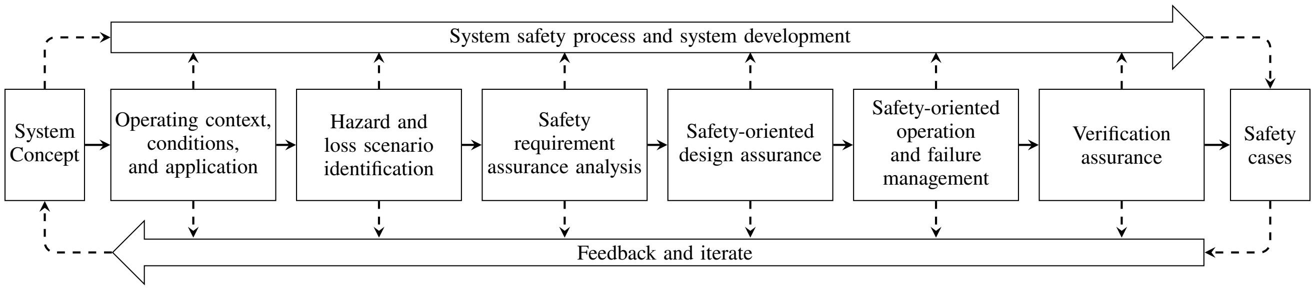


Fig. 3 Overview of wearable robotics exoskeleton systems assurance, adopted from Hawkins et al [92]

- Have mechanical components undergone routine maintenance and inspections?
- Is the exoskeleton in compliance with safety requirements and regulations?
- Have users undergone comprehensive training on proper exoskeleton operation?
- Are users aware of potential risks and safety measures during operation?
- Is there a feedback mechanism for users to report observed abnormalities?
- Is there a clear emergency shutdown mechanism in place?
- Have users been trained on emergency response procedures?
- Are there protocols for addressing unforeseen motions or unprecedented HRI?
- Is there a real-time monitoring system for continuous assessment of the exoskeleton's performance?
- Does the system have the capability to detect and respond to sensor failures or communication breakdowns?
- Are there mechanisms in place for real-time adjustments based on user feedback and environmental conditions?
- How is the exoskeleton designed to safely interact with the user's movements seamlessly?
- Have predictive simulation tests been conducted to ensure user comfort and prevent unexpected motions?
- Are there provisions for user feedback and adjustments during operation to enhance comfort and usability?

4 Case studies and experiments

Herein, we present two successful case studies that formulated and validated an exoskeleton design using a simulation framework (with regard to HRI) and HITL experimental evaluations.

4.1 Shoulder active-passive exoskeleton. In the initial phase of the research, a shoulder active-passive exoskeleton design underwent a comprehensive multibody dynamic simulation (Figure 4(a)). Subsequently, the designed exoskeleton was tested in the second phase through a HITL experiment.

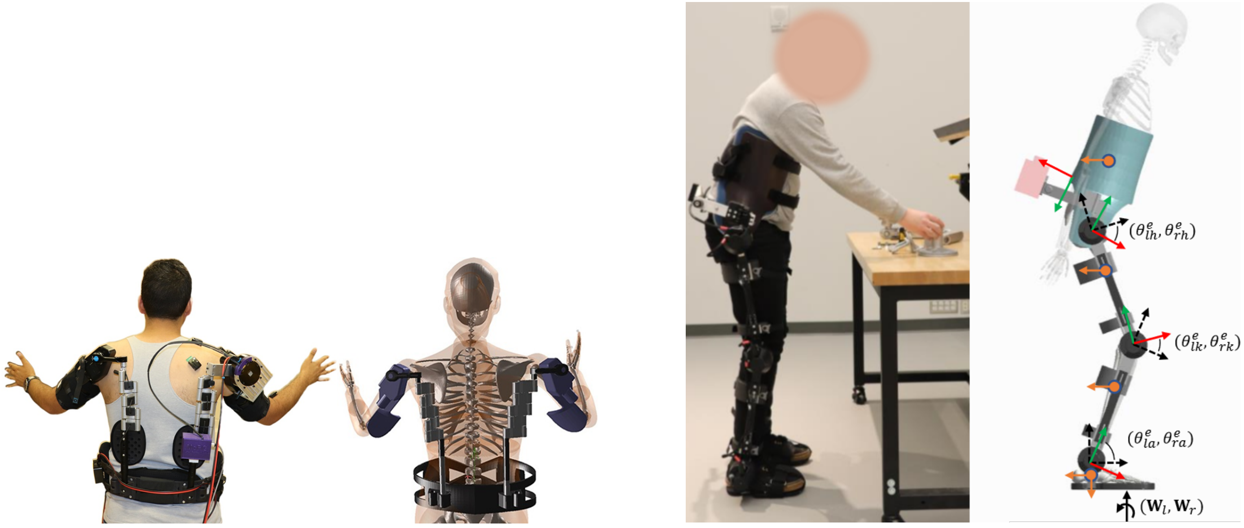
During the design phase, a multibody dynamic model was developed, encompassing six key components: an upper-body MSK model [26], an optimal controller [81], the rigid bodies of the exoskeleton [35], a passive mechanism [93], a powered actuator, and an assistance model [81]. The design process involved experimental measurements derived from six distinct tasks. Optimization of the system safe design was achieved by strategically selecting features of the passive mechanism and exoskeleton motor to minimize human joint active torque, power, muscle metabolic energy expenditure, and actuator electricity consumption.

Moving on to the second phase, a test framework was developed for evaluating the shoulder exoskeleton. Additionally, a sophisticated sEMG-based human-robot cooperative control method was devised to execute the wearer's movement intentions [73]. The hierarchical control structure incorporated sEMG-based intention estimation, mid-level strength regulation [81], and low-level actuator control. This control paradigm was then applied to the shoulder joint elevation experiments to assess the safety and efficacy of the exoskeleton controller. A comparative analysis was conducted between the active-passive assistance and fully-passive and fully-active exoskeleton controls, employing criteria such as post-test surveys, load tolerance duration, and computed human torque, power, and metabolic energy expenditure based on sEMG signals and inverse dynamic simulation.

The exoskeleton is equipped with a number of safety features and enhancements, including: I) using sturdy engineering-standard plastic for the armrests; II) a frame with 4 lift adjustments and an accordion mechanism to dynamically conform to the intricacies of the shoulder joint; III) use of a 24-volt battery-powered actuator, instead of relying on an electric grid, along with the inclusion of two emergency shut down switches for power source disconnection; IV) implementation of a parallel actuator motor with a passive mechanism to increase power and reduce actuator weight and torques [35]; V) limitation of the actuator's maximum torque within the on-board driver through indirect measurement of output torque, as well as the use of saturation in the mid-level controller [81]; VI) the robust industrial standard CAN connection has been used between controller brain and the actuator driver; VII) deployment of a robust and predictive algorithm to compensate for lack of input data (in case of sEMG sensor disconnection) and measurement delay [73]; VIII) use of a machine learning model to interpret the sEMG signal, ensuring consistency and limiting the output to the maximum possible value for predicted torque and joint angle; and IX) the gain of controllers is gradually increased if the user has not adapted to the exoskeleton power assist feature [81].

4.2 Lower-limb standing/walking exoskeleton. Similar to the shoulder active-passive exoskeleton in section 4.1, testing with the lower-limb exoskeleton first began with multibody dynamic simulations (Figure 4(b)). The focus of these simulations pertained to the maintenance of upright balance [61]. Often, it is difficult for end-users to remain upright and stable when wearing a lower-limb assistive-device without assistance from the device itself or crutches [48,59,60].

The multibody dynamic model developed consisted of the following two subsystems merged through kinematic constraints: i) a whole-body MSK model and ii) the actuated structure of the lower-limb exoskeleton. As the device referenced during the design process was the Technaid EXO-H3, actuation and DoFs were limited to the sagittal plane. This assumption allowed focus to be placed on both walking and standing without concern over lateral stability; however, one should acknowledge the importance of such features in the design of fully autonomous assistive devices, e.g., those that can automate walking for paraplegic users [59]. In the human model, joints were actuated via computationally-efficient muscle torque generators [26,27,94,95], while any of the hip-knee-ankle robot joints could be actuated through a simple "lossy" torque generator with reflected mass/inertia. In designing this model and using it in simulations prior to experiments, feasibility of a sagittal balance recovery following a perturbation could be investigated. For example, the efficacy of assisted feet-in-place balance recovery could be assessed given different levels of device actuator output



(a) Shoulder active-passive exoskeleton (EVO, Ekso Bionics)

(b) Lower-limb standing/walking exoskeleton (EXO-H3, Technaid)

Fig. 4 Illustration depicting two instances of successful safety validation observed through multibody dynamic simulations and human-in-the-loop (HITL) experiments

saturation or in the case in which the end-user must provide all torques via muscle activations [61,96]. Due to hardware and/or processing limitations, it can also be difficult to devise systems to estimate user intent, e.g., limited sEMG electrode placement with device and onboard sensors lacking fidelity or resolution. In these cases, simulations can also support the researcher(s) by providing a tool with which they can assess sensitivity of standing/walking performance when the end-user actions and exoskeleton control law do not fully align, e.g., minimizing different performance criteria. For further details on the model and dynamic simulations, refer to Inkol and McPhee [61].

The active lower-limb exoskeleton model was designed according to the topology of the Technaid EXO-H3. The device itself (total mass $\approx$ 16.5 kg can be reduced to a series of high-output electric actuators connected by lightweight, rigid linkages. Actuators consist of BLDC motors with high-ratio strain-wave gearing. No modifications were made by the research team on the device at the time of writing this article. Technaid includes a variety of onboard safety features with their system, both to protect the end-user and device hardware:

- Limits placed on torque generation when using motor torque control ($\pm$35 N m at output shaft);
- Motors disengage at device end-RoMs;
- Actuators are covered with plastic housings that insulate and fireproof the onboard electronics. Also prevents fingers from being trapped near the output shaft.

Other considerations that Technaid made with regards to safety include preventing pregnant individuals from donning the device, enforcing height/weight requirements, and assessment of whether the user is currently dealing with a MSK or skin injury as well as limited passive RoM. The device itself interacts with the end-user via elastic strapping and cuffs that use soft materials where possible. High pressure applications are possible so one must be aware of broken/damaged skin, sensitivities, etc. Onboard applied torque sensors are also available to infer the interaction forces between device and end-user; such sensors could be used to modify control laws such that they better follow human-intention and thus avoid high contact pressure. Lastly, given the focus of our research, we must consider safety of balance recovery versus the safety of torque application. A fall could be catastrophic for both device and end-user health. In this sense, manufacturers often mandate end-users to utilize additional devices for balance regulation that can limit arm movement, e.g., a walker or crutches.

In recent experiments, we were interested in safe crutch-less stance while donning a rigid lower-limb exoskeleton (the EXO-H3), i.e., could the end-user remain upright and balanced in the EXO-H3 following a perturbation as suggested by simulations and experiments [48,60,61,97]. All testing was reviewed and received ethics clearance through a University of Waterloo Research Ethics Committee. Initial testing involved primarily feedforward impedance control. Aside from the EXO-H3's onboard safety features, actuator torques were also saturated in the control test bench software (MATLAB). Additionally, data from the onboard kinematic sensors (joint position/velocity) was used in real-time to estimate when the participant was in a bilaterally symmetric standing posture (state), i.e., when it would be safe to implement feet-in-place assistance like in Emmens et al [48]. Lower-limb exoskeleton testing took place with healthy young adult participants exclusively; testing with clinical or aging populations can require more intensive monitoring of participant state, e.g., whether they they have any or are developing fragile skin on their legs (where the exoskeleton must apply forces). To combat user fatigue, rest was offered and provided appropriately. Rest is especially important in these protocols depending on how the control architecture is structured, e.g., not compensating for gravitational forces resulting from device inertia could increase the rate of developing fatigue and subsequent injury. Testing also involved each participant harnessed to ceiling-mounted track. A load cell was included in series in order to determine whether a fall would have occurred without a harness, i.e., assess whether the participant relied on the harness to stay upright. After the final trial, participants were asked whether they felt any discomfort while wearing the device via the local discomfort body region heat maps used by Meyer et al [98]. These maps could then be used to improve participant safety and comfort in future protocols.

5 Future Directions

Safety perception was quantified using a 9-point scale, ranging from -4 (depicted by a sad smiley face) to +4 (indicated by a happy smiley face), with 0 representing a neutral expression [99]. In the future development of exoskeleton technology, advancing towards more robust and standardized safety frameworks will be essential. One critical area of improvement lies in the establishment of scoring metrics that can quantitatively assess the design and safety capabilities of exoskeleton systems across various applications, usage scenarios, environments, and user demographics. These scoring metrics should account for specific needs and challenges inherent to different user groups, such as elderly individuals, patients with mobility impairments, or industrial workers. Moreover, defining safety regions fitted to each application context—whether in medical rehabilitation, industrial assistance, or daily mobility support—will be crucial. Such safety regions would consider factors like movement dynamics, task requirements, environmental conditions, and the physical capabilities of users. Emphasizing these aspects will not only enhance the safety and effectiveness of exoskeleton technology but also streamline system iteration and regulatory compliance processes, ultimately facilitating broader adoption and integration into real-world settings.

6 Conclusions

In conclusion, this paper systematically addressed a spectrum of challenges associated with exoskeleton technology, spanning mechanical, regulatory, and ethical domains. By delineating potential risks and proposing practical solutions rooted in design, control, and testing strategies for each identified challenge, the paper aims to provide a comprehensive guideline for developers and users. Emphasizing safety, reliability, and optimal performance, the guideline covers pre-operation checks, user training, emergency response, real-time monitoring, and user interaction. Through the exploration of two case studies and experiments, the paper illustrates successful implementations of safety measures. As the field of exoskeleton technology continues to evolve, the insights presented in this paper contribute to the responsible innovation and user-centric development of these wearable robotic devices across various applications, from healthcare to industrial settings. The outlined framework and recommendations serve as guidance for stakeholders navigating the intricacies of exoskeleton design, ensuring a balance between technological advancement and user safety.

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Abbreviations

BLDC brushless direct current.

CAN controller area network.

CNS central nervous system.

DoF degree of freedom.

HITL human-in-the-loop.

HRI human-robot interaction.

IMU inertial measurement unit.

MSK musculoskeletal.

RoM range of motion.

sEMG surface electromyography.

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